Market Risk Monitoring and Early Warning Based on Regime Classification

Catalog

1. Problem Setting

Financial markets occasionally transition from stable periods into stressed environments characterized by rapid price declines, elevated volatility, and deterioration in credit conditions. These transitions are difficult to anticipate using simple indicators.

This project studies whether multi-asset market data can be used to anticipate future downside risk events in equity markets. The analysis integrates signals from equities, volatility indices, interest rates, foreign exchange, and credit markets.

The project combines both supervised and unsupervised learning approaches. An unsupervised regime detection step summarizes overall market conditions, while a supervised learning model predicts future downside risk events using historical data.

2. Problem Definition

The main objective of this project is to predict whether the market will experience a significant downside risk event in the near future.

A supervised learning framework is used to forecast future risk outcomes based on historical market indicators. The prediction is made using information available up to time t, while the outcome corresponds to future market behavior over a predefined horizon.

Market regimes are inferred using unsupervised learning methods and are used as additional explanatory variables. These regimes summarize recurring patterns of market conditions such as calm periods, volatile stress periods, or transitional phases.

3. Data Sources

Data for this project is obtained from multiple financial data providers. Primary sources include:

• Yahoo Finance historical market data

• Interactive Brokers (IBKR) market data

The datasets include equity indices, volatility indicators, interest rate series, foreign exchange rates, and credit market measures.

4. Data Description

The dataset is a daily multi-asset financial panel covering global equities, volatility indices, ETF of US government bond, ETF on credit, commodities, and foreign exchange markets from 1980 to 2026. Price data are sourced from Yahoo Finance (adjusted for splits and dividends), while USD/CNH data are obtained from Interactive Brokers using midpoint daily bars. The dataset includes OHLC fields. The data are structured as a MultiIndex panel (price type × asset) indexed by trading date, suitable for macro-financial modeling, risk estimation, and multi-asset machine learning applications.
Data description is as following table:

Category	Ticker	Symbol	Description	Start Date	End Date
Equity	EEM	EEM	iShares MSCI Emerging Markets ETF	2003-04-14	2026-03-04
Equity	EUROSTOXX50	^STOXX50E	Euro Stoxx 50 Index	2007-03-30	2026-03-04
Equity	NASDAQ100	^NDX	NASDAQ 100 Index	1985-10-01	2026-03-04
Equity	NIKKEI225	^N225	Nikkei 225 Index (Japan)	1980-01-04	2026-03-04
Equity	SP500	^GSPC	S&P 500 Index	1980-01-02	2026-03-04
Volatility	VIX	^VIX	CBOE Volatility Index	1990-01-02	2026-03-04
Interest Rates	IEF	IEF	iShares 7-10 Year Treasury Bond ETF	2002-07-30	2026-03-04
Interest Rates	RATE_10Y	^TNX	10-Year Treasury Note Yield	1980-01-02	2026-03-04
Interest Rates	RATE_3M	^IRX	3-Month Treasury Bill Rate	1980-01-02	2026-03-04
FX	DXY	DX-Y.NYB	US Dollar Index	1980-01-02	2026-03-04
FX	EURUSD	EURUSD=X	Euro/US Dollar Exchange Rate	2003-12-01	2026-03-04
FX	USDCNH	USD.CNH	US Dollar/Chinese Yuan Offshore Exchange Rate (from IBKR)	2012-10-03	2026-02-11
FX	USDJPY	USDJPY=X	US Dollar/Japanese Yen Exchange Rate	1996-10-30	2026-03-04
Credit	HYG	HYG	iShares iBoxx High Yield Corporate Bond ETF	2007-04-11	2026-03-04
Commodities	COMMODITY	^SPGSCI	S&P GSCI Commodity Index	1984-01-03	2026-03-04

Table 1 Data Description

5. Data Exploration

Exploratory data analysis (EDA) is performed to understand the statistical properties of the dataset and to identify patterns relevant for predictive modeling.
Exploration methods include:

• Time series visualization

• Distribution analysis
• Correlation analysis
• Rolling statistics
• Cross-asset relationships

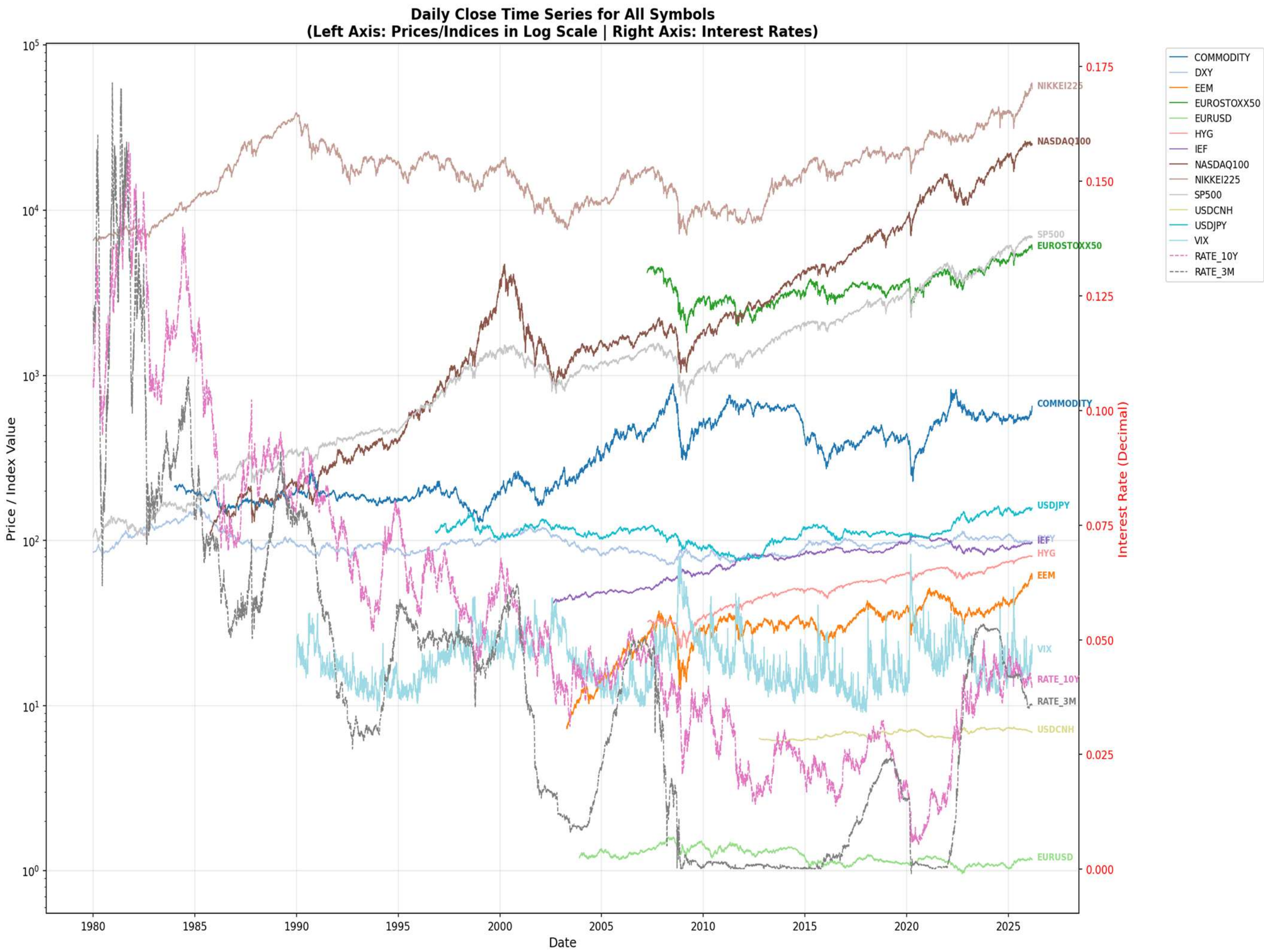

Figure 1 Time series overview of key market indicators

Figure 1 shows the long-term time series of major market indicators used in the analysis, including equity indices, volatility indicators, interest rates, credit market ETFs, and foreign exchange variables. The figure illustrates the dynamic evolution of global financial markets from 1980 to 2026. Several major financial crises are clearly visible in the time series, including the 2008 global financial crisis, the COVID-19 market crash in 2020, and the tightening cycle beginning in 2022. These periods are characterized by sharp market declines and elevated volatility, highlighting the importance of identifying early warning signals for systemic market stress.

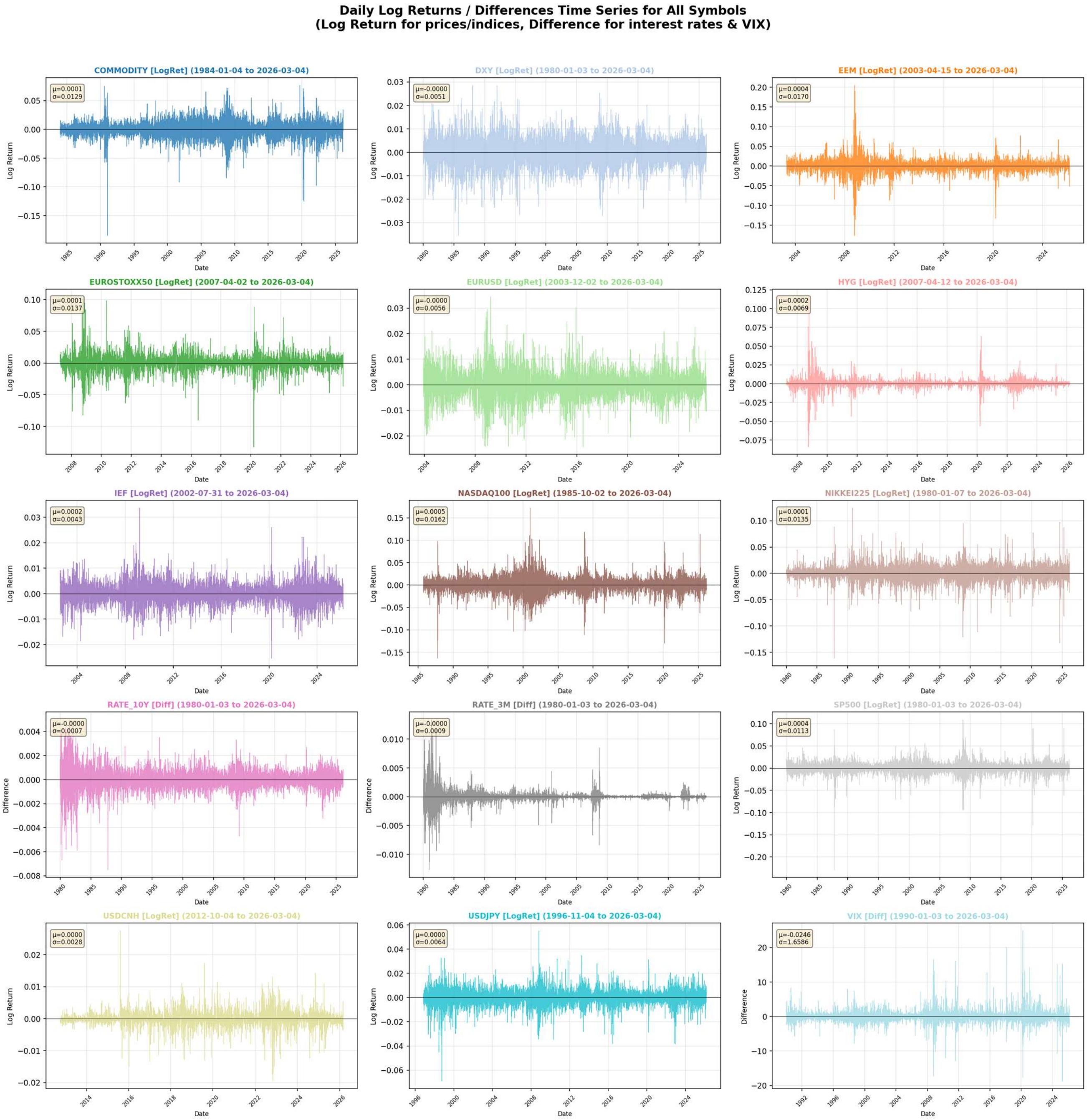

Figure 2 Daily Log Return of key market indicators

Figure 2 displays the daily logarithmic returns (or difference) of selected market indicators. Log returns are used instead of raw prices because they provide a stationary representation of asset dynamics and allow better comparison across assets with different price levels. The return series exhibit clear volatility clustering, where large market movements tend to occur in groups. This phenomenon is particularly visible during crisis periods. The presence of volatility clustering suggests that past market behavior contains predictive information about future market risk.

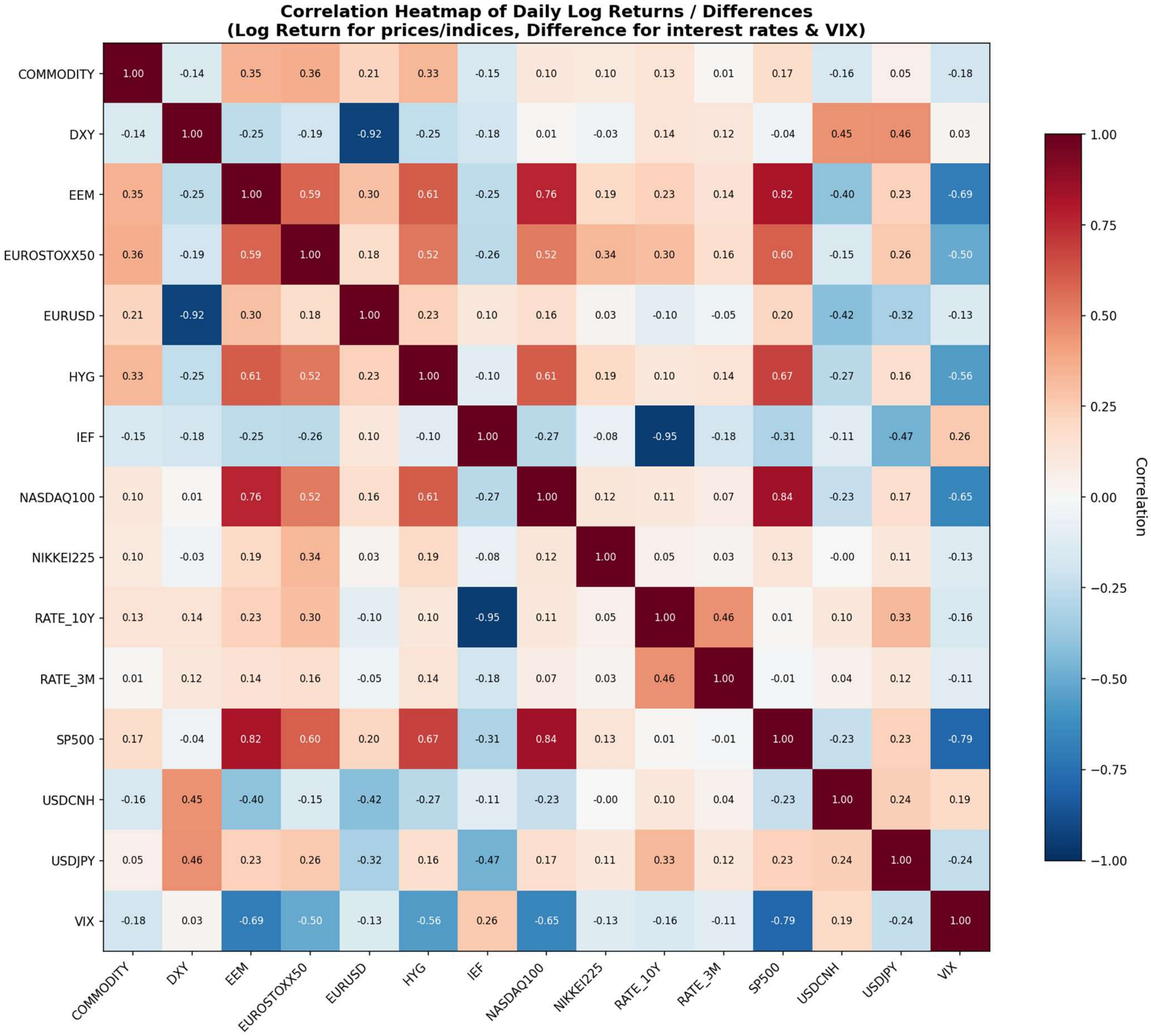

Figure 3 Correlation Heatmap of Market Indicators

Figure 3 presents the correlation matrix of key financial indicators used in the analysis. The heatmap highlights the strength and direction of pairwise relationships among equities, volatility indices, credit markets, and macroeconomic variables. Several economically meaningful relationships emerge. Equity indices tend to be highly correlated with each other, while volatility indicators often display negative correlations with equity returns. Credit market indicators and equity indices also exhibit strong co-movement during stress periods, indicating the presence of systemic risk channels across financial markets.

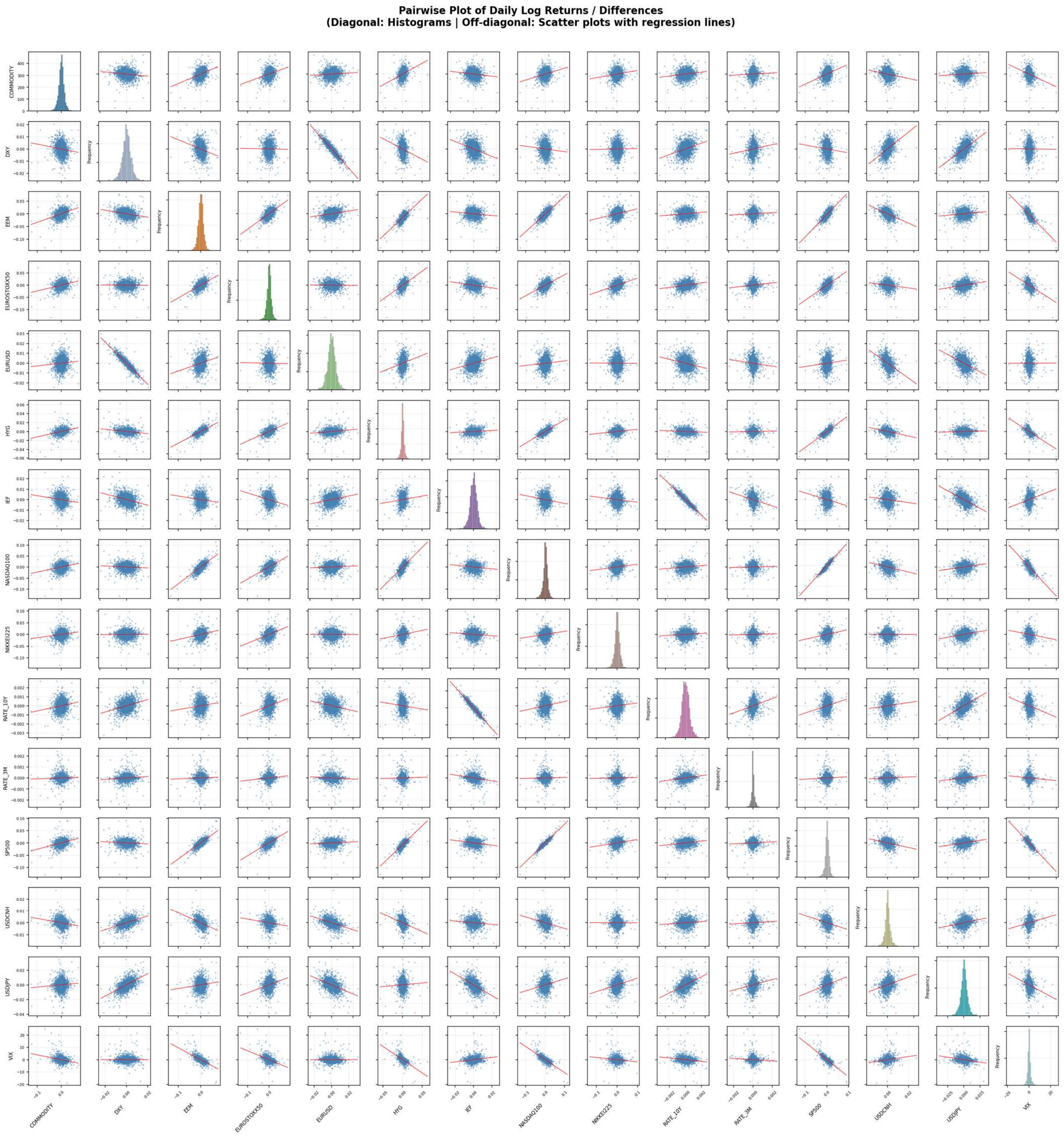

Figure 4 Pairwise Plot of Market Indicators

Figure 4 shows pairwise scatter plots of selected financial indicators. These plots help visualize nonlinear relationships and interactions among key variables. The pairwise relationships reveal potential nonlinear patterns between variables, suggesting that simple linear models may not fully capture market dynamics. This observation motivates the use of machine learning models capable of modeling nonlinear relationships in the prediction of market risk.

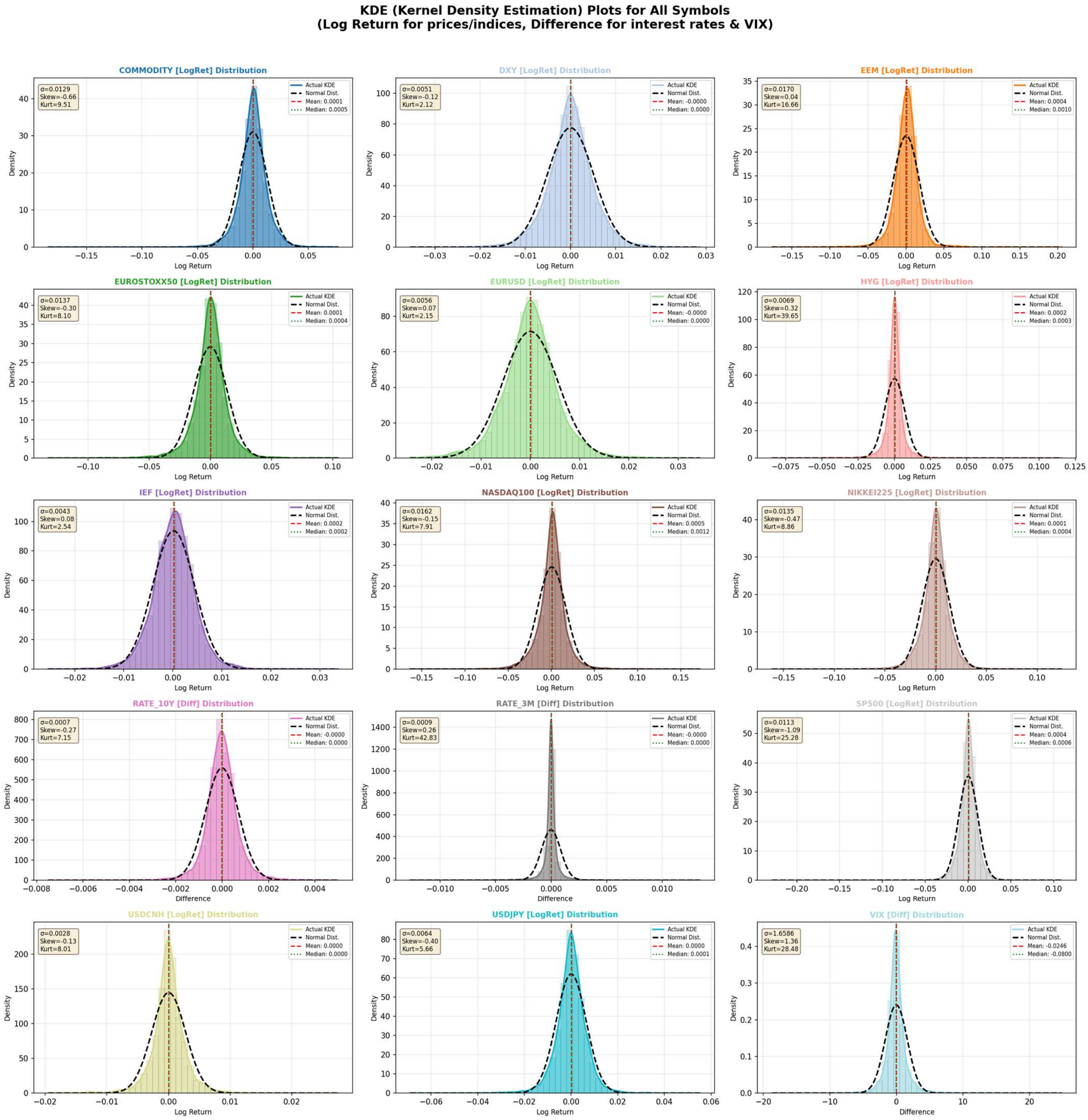

Figure 5 Kernel Density Estimation of Market Indicators

Figure 5 displays kernel density estimations of the distributions of selected financial indicators. The distributions show clear departures from normality, with heavy tails and skewness in several variables. These characteristics are common in financial time series and highlight the importance of robust modeling techniques capable of capturing extreme market movements.

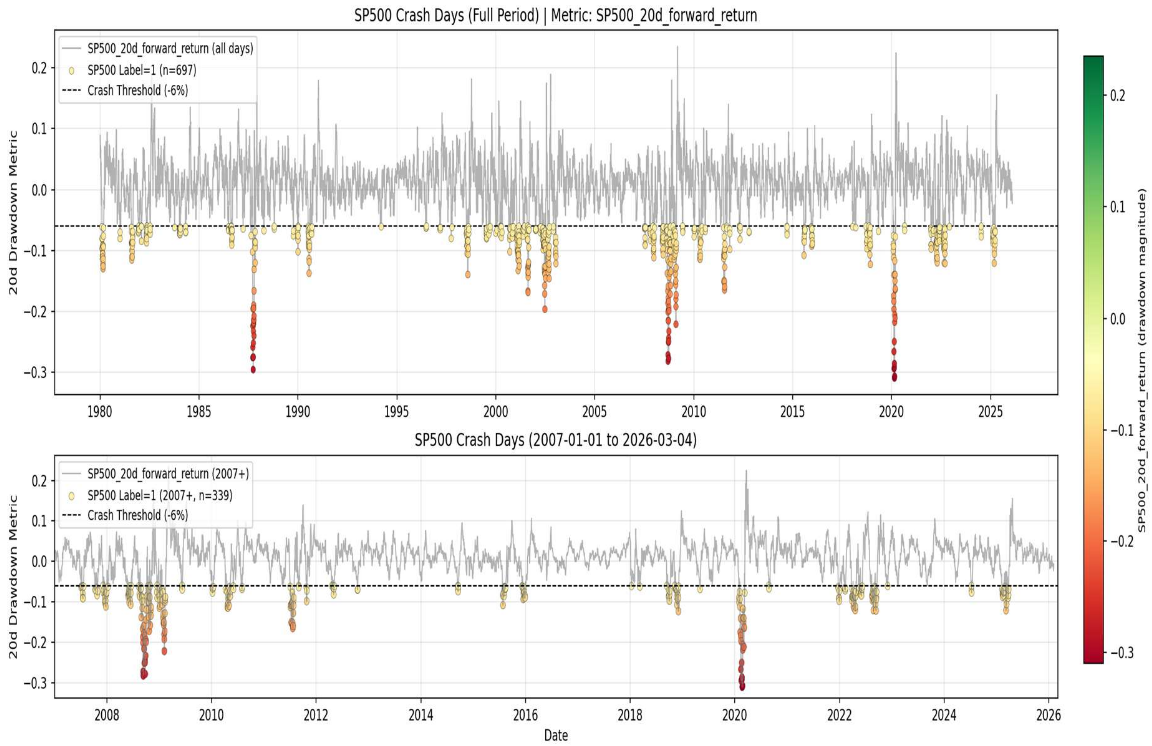

Figure 6 Historical Market Crash Days

Figure 6 illustrates the historical occurrence of market crash days based on the defined crash threshold. In this study, a market crash is defined as a maximum drawdown greater than 6% within the subsequent 20 trading days for major market indices.

Crash events are relatively rare but tend to cluster in time during periods of systemic stress. This clustering behavior further supports the hypothesis that market crashes are influenced by underlying structural conditions in financial markets.

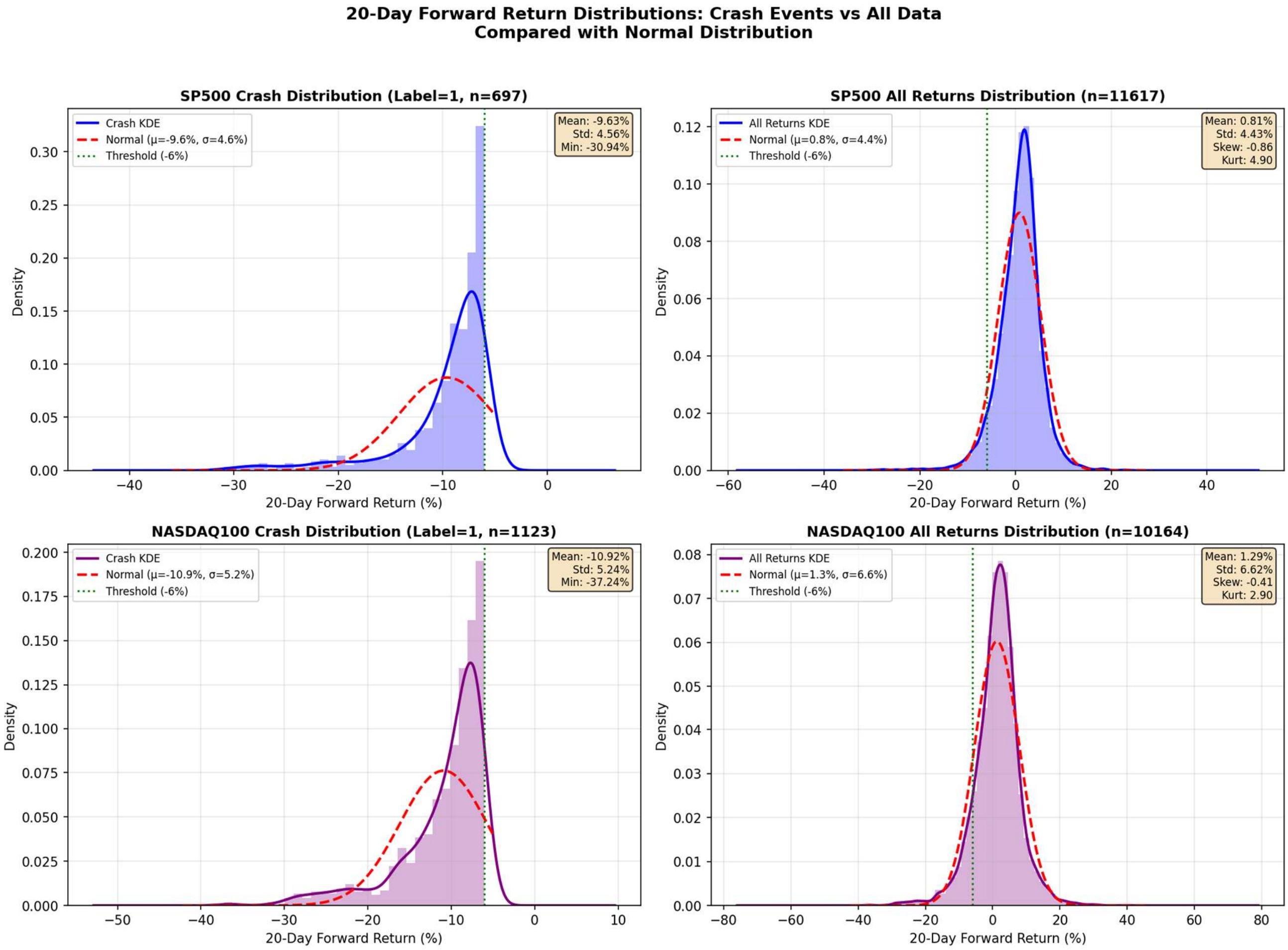

Figure 7 Return Distribution of Crash Days V.S. of All Days

Figure 7 compares the distribution of daily returns during crash days with the distribution of returns across all trading days. Crash days exhibit significantly heavier negative tails compared with the overall return distribution. This difference confirms that crash periods represent extreme market events and justifies the use of classification models to predict such rare but impactful outcomes.

6. Data Mining Tasks

Several data mining tasks are performed in this project, including:

- Feature engineering
- market regime detection
- Dimensionality reduction and feature selection
- Supervised classification for risk prediction

Feature engineering

To improve the predictive performance of crash forecasting while preserving economic interpretability, this project constructs a multi-layer feature set from the raw market panel. The feature engineering process is designed to capture short-term shocks, medium-term persistence, long-term tail behavior, cross-asset interactions, and latent market stress states. The final feature matrix includes transformed return variables, volatility-related features, credit-spread features, rolling statistics, regime-aware interaction terms, and recurrence-based nonlinear descriptors.

Data alignment and missing-value handling

The raw input is a MultiIndex market panel containing OHLC fields for multiple assets. Before constructing features, all series are aligned using S&P 500 close prices as the baseline trading calendar. Any date for which SP500 close is missing is removed. For all other symbols, missing observations after the first valid point are filled using forward fill, so that the panel remains synchronized across assets without artificially creating pre-inception history. This design ensures that the equity market benchmark defines the usable timeline while preserving as much cross-asset information as possible.

Basic return transformations

Because different variables have different economic meanings, the project does not apply a single transformation to all series. For price-like assets such as equity indices, ETFs, commodities, and FX rates, the daily log return is used. For level variables such as VIX, 10-year Treasury yield, and 3-month rate, the daily first difference is used instead. This distinction is important because yields and VIX are already level indicators rather than tradable prices, so differencing preserves their economic interpretation better than percentage return scaling.

Intraday range features

For each asset, an additional intraday volatility proxy is constructed using the logarithm of the high-low ratio. This feature measures the size of the intraday trading range and serves as a simple but effective proxy for market uncertainty or realized intraday stress. The code also performs data-quality checks: if an invalid case with $High < Low$ appears, the range is set to zero; extremely large daily ranges are capped to avoid distortion from obvious bad prints or recording errors.

Yield-curve slope

To capture macro-financial conditions, the project constructs the **term spread** between the 10-year and 3-month rates. This feature summarizes yield-curve steepness and helps represent monetary tightening, inversion, and changes in macro expectations that may affect future market stress.

S&P 500 realized volatility and rolling maximum drawdown

Two important market-state variables are extracted directly from SP500 behavior. First, the project computes 20-day rolling annualized realized volatility from SP500 log returns. Second, it computes 20-day rolling maximum drawdown, which captures the most severe recent cumulative decline inside a rolling window. This variable is especially useful in crash prediction because it summarizes whether the market is already entering a stressed downside trajectory. Both measures use flexible min periods settings so that early observations are not discarded unnecessarily.

Credit factor

A key cross-asset stress feature is the credit factor, defined as the daily excess return of high-yield bonds over Treasury bonds. This feature is intended to proxy changes in credit conditions. When credit spreads widen and risk appetite deteriorates, high-yield bonds typically underperform Treasury bonds, causing the credit factor to weaken. Therefore, this variable acts as a compact market-based measure of credit stress.

Rolling aggregates and long-horizon tail descriptors

To capture temporal persistence, the project augments the base features with rolling summaries. For each base feature, it constructs: a 5-day rolling mean, a 10-day rolling mean and a 252-day rolling 90th percentile.

The 5-day and 10-day averages summarize short-term and medium-term trend or persistence, while the 252-day 90th percentile provides a long-horizon tail benchmark that helps encode whether a variable is currently elevated relative to its recent one-year distribution. For VIX-related rolling features, the code uses the VIX price level itself instead of VIX daily difference, because the level carries more direct economic meaning for market fear.

Regime-aware stress features

Because the project includes a regime-classification layer, feature engineering is extended to incorporate information from the estimated stress regime probability P_Stress. In particular, it uses Delta_P_Stress (Change of P_Stress), its 5-day rolling mean, and a 252-day z-score of that smoothed series to capture the recent acceleration of stress probability. This transformation is valuable because it focuses not only on the level of stress, but also on whether stress is increasing unusually quickly relative to its historical pattern.

Cross-asset interaction features

To explicitly model nonlinear stress propagation across markets, several interaction terms are constructed. The first is SPREAD_VIX_HIGH, which multiplies a credit-stress gate and a VIX-stress gate. Each gate maps the underlying variable into the $[0, 1]$range using rolling 252-day 80th and 95th percentile thresholds. This feature becomes large only when both credit stress and volatility stress are simultaneously elevated, capturing joint systemic-risk

conditions. The second is SPX_VIX_SPIKE, **t**his term is designed to detect equity selloffs accompanied by upward volatility shocks, which are typical characteristics of market panic episodes. The third is DXY_SPREAD_INTERACT, defined as the product of 20-day cumulative dollar return and the z-score of the credit factor. This feature aims to capture joint movement between USD strength and credit-market deterioration, which often appears during global risk-off episodes.
The project also constructs a set of interaction terms between regime probability and market features, including:

- VIX_VOL_RATIO = VIX / SPX_VOL_20D, which measures implied volatility relative to realized volatility
- SPX_VOL_STRESS, defined as the z-score of SPX realized volatility multiplied by P_Stress
- HYG_RANGE_STRESS, defined as the z-score of HYG short-term range multiplied by P_Stress
- NASDAQ_RANGE_STRESS, defined as the z-score of NASDAQ100 intraday range multiplied by P_Stress

These features are intended to amplify market signals only when the latent regime model indicates elevated stress, thereby improving the model's sensitivity to regime-dependent nonlinearities.

Recurrence-based nonlinear features

A distinctive part of this project is the introduction of recurrence plot statistics. For selected series — specifically SP500 log return, VIX difference, credit factor, and DXY log return — the code computes two recurrence-quantification-analysis (RQA) metrics over a rolling 60-day window:

- Recurrence Rate (RR): measures how often the system revisits similar states
- Determinism (DET): measures how much of the recurrence structure forms diagonal-line patterns, indicating repeated temporal structure

The recurrence matrix is built using a threshold 0.5 std of the windowed series. These features are useful because conventional rolling means or volatilities only summarize first- and second-moment behavior, while RR and DET attempt to capture structural persistence, pattern repetition, and nonlinear temporal organization in the market. This is especially relevant when markets transition between calm, transitional, and stressed regimes.

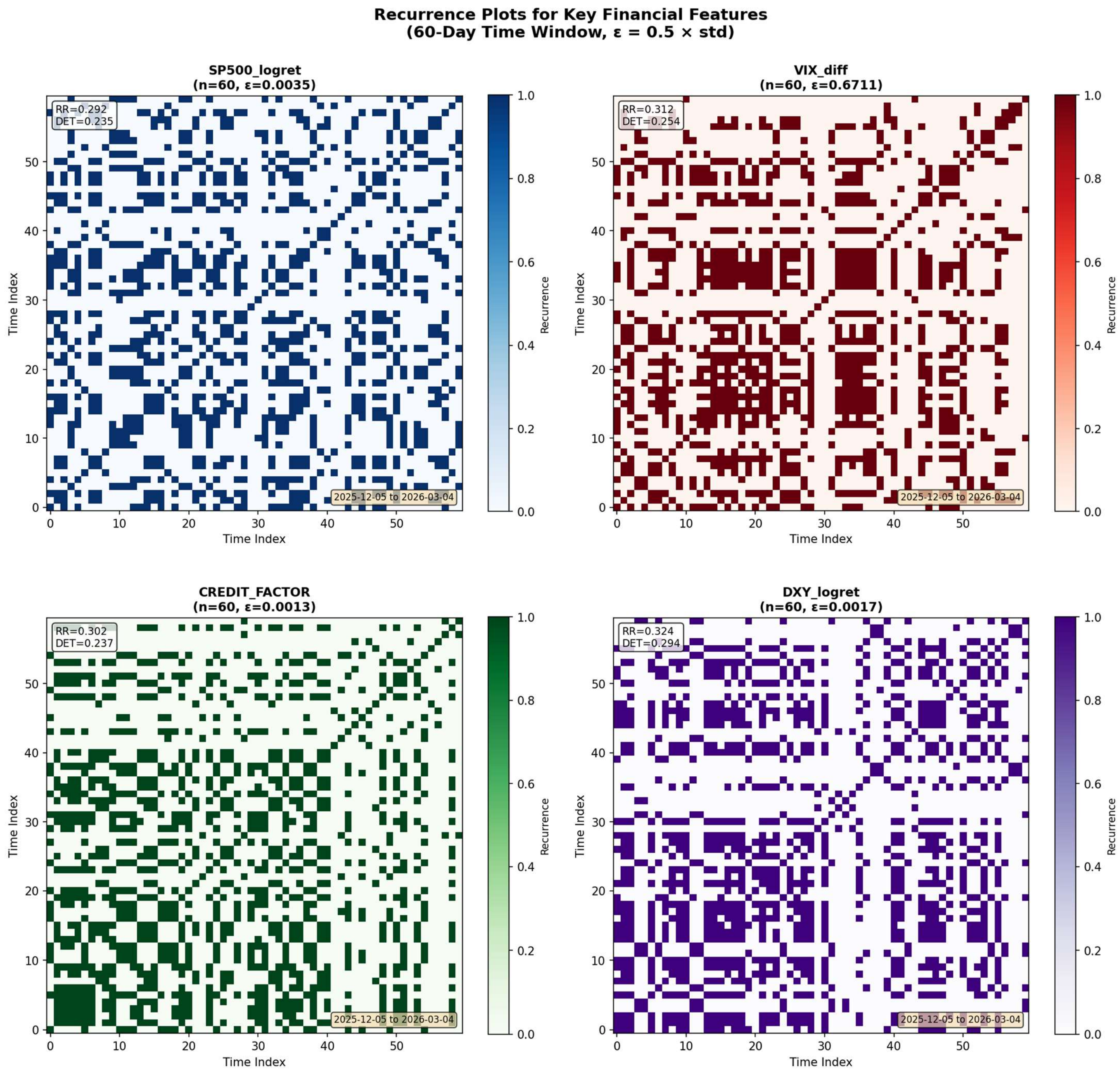

Figure 8 Recurrence Plots of Key Financial Features

Market regime detection

To capture latent market conditions and improve the predictive power of the crash-prediction model, this project introduces a market regime detection layer based on a Hidden Markov Model (HMM). The purpose of regime detection is to identify recurring macro-financial states of the market, such as calm environments, transitional periods, and stress regimes. These regime signals are then incorporated into the feature set as additional explanatory variables for the supervised prediction model.

Input Variables for Regime Detection

The regime model uses four macro-financial indicators that capture different channels of systemic market risk:

- SPX_VOL_20D: 20-day realized volatility of the S&P 500
- VIX Close: the level of the VIX volatility index
- CREDIT_FACTOR: the credit spread factor defined as the return difference between high-yield bonds (HYG) and Treasury bonds (IEF)
- RATE_SLOPE_10Y_3M: the yield curve slope defined as the difference between the 10-year and 3-month Treasury yields

These variables represent the main dimensions of macro-financial risk

Rolling Z-Score Standardization

Before fitting the HMM, each variable is standardized using a 250-day rolling z-score. This rolling standardization serves two purposes: Scale normalization – ensuring that variables with different units can be used jointly in the HMM; Leakage prevention – the rolling statistics are computed using only historical data up to time t, preventing the use of future information. A minimum of 200 observations is required before the rolling statistics become valid.

Hidden Markov Model Specification

The regime detection model uses a Gaussian Hidden Markov Model with three hidden states. Each hidden state represents a different market environment characterized by distinct volatility, credit, and macro conditions

Rolling Window Estimation

Instead of fitting the HMM once using the entire dataset, the model uses a rolling window estimation framework to reflect changing market dynamics. This means: The HMM is estimated using the most recent ~5 years of data; The model is re-estimated approximately once per month.

Rolling estimation allows the regime model to adapt to evolving financial structures, such as changes in volatility regimes or macroeconomic conditions

Identification of Regime Labels

After fitting the HMM, the model assigns an economic interpretation to each hidden state based on the average VIX level observed within that state: states are ranked according to their mean VIX z-score. This approach provides an economically intuitive interpretation of the hidden states and ensures consistency across rolling estimation windows.

Forward-Only Probability Estimation

A key methodological feature of this project is the use of a forward-only filtering algorithm to compute regime probabilities. Unlike the standard HMM forward-backward algorithm, which uses the full sample including future observations, the forward-only algorithm calculates the state probability using only information available up to time t. This prevents look-ahead bias and ensures that the regime signal can be used in a realistic forecasting setting

Backfilling the Initial Period

Because the rolling HMM requires an initial estimation window of 1200 observations, the first part of the dataset does not immediately have regime estimates. To maintain a complete time series, the project performs a backfilling procedure:

1. The first successfully estimated HMM model is stored.
2. The forward algorithm is applied retrospectively to the earlier observations using this model.

This ensures that regime probabilities are available for the entire historical dataset.

Regime Probability Outputs

For each trading day, the regime model produces: $P(\text{Calm})$, $P(\text{Transitional})$ and $P(\text{Stress})$
These probabilities are used instead of discrete regime labels because they provide smoother and more informative signals. Additionally, the project computes daily change in stress probability and serves as a useful indicator of rapidly increasing market risk

Role of Regime Detection in the Prediction Framework

The regime detection component serves three important roles in the overall modeling pipeline:

1. Market state summarization: It provides a compact representation of the complex financial environment.
2. Feature generation: Regime probabilities and regime-dependent transformations are added to the feature set.
3. Risk interpretation: The regime model helps explain how predicted crash probabilities relate to underlying market states.

By combining regime classification with supervised learning, the project aims to capture both structural market dynamics and short-term predictive signals, leading to more robust crash prediction.

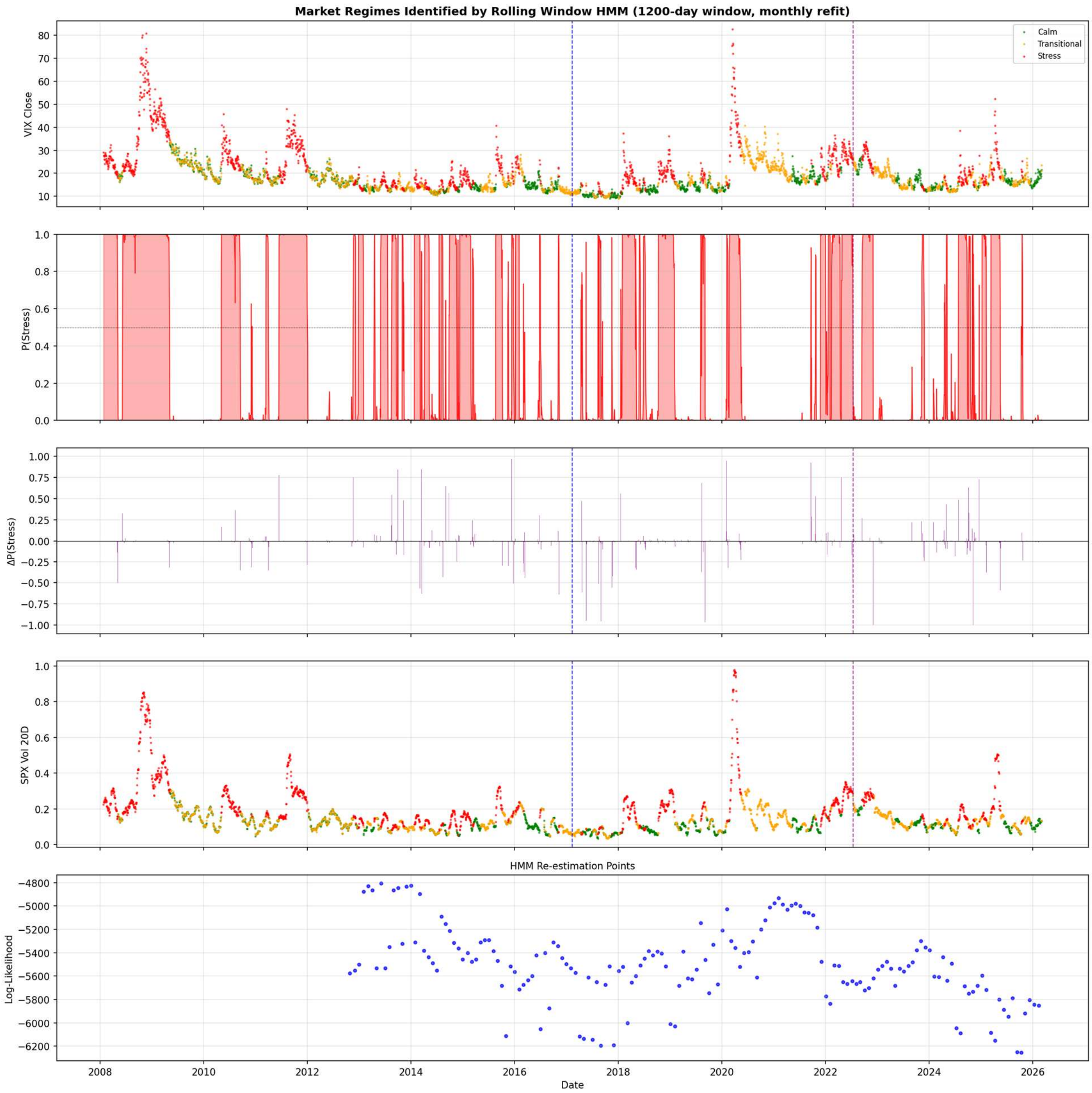

Figure 9 Market Regimes Identified by Rolling Window HMM

Figure 9 presents the market regimes identified by the rolling-window Hidden Markov Model (HMM) using a 1200-day estimation window with periodic refitting. The figure combines several layers of information, including the VIX level, the estimated probability of the stress regime, changes in the stress probability, the SPX 20-day volatility, and the model log-likelihood over time. Together, these panels provide a dynamic view of how latent market states evolve across different macro-financial environments.

The top panel shows that periods classified as Stress are generally associated with elevated VIX levels, while Calm periods tend to coincide with lower and more stable volatility. Transitional periods usually appear between these two extremes, indicating that the model does not force market conditions into a purely binary calm-versus-crisis structure. This is important because many market environments are neither fully tranquil nor fully stressed, but rather in the process of shifting from one state to another.

The second panel, which plots the estimated probability of the stress regime, shows that the model is able to identify episodes of persistently high market stress. Notable stress periods appear around the 2008 global financial crisis, the 2020 COVID-19 shock, and several later volatility episodes. These intervals are characterized by high and sustained values of $P(\text{Stress})$, indicating that the HMM successfully captures prolonged periods of systemic tension rather than isolated daily fluctuations.

The third panel shows changes in the stress probability, $\Delta P(\text{Stress})$, which can be interpreted as a regime-transition signal. Large positive spikes indicate rapid increases in market stress, while negative spikes reflect easing conditions. This panel is particularly useful for early warning purposes, because abrupt increases in stress probability may precede or accompany important regime shifts in financial markets.

The fourth panel shows SPX 20-day volatility colored by the inferred regime labels. The consistency between elevated realized volatility and the stress regime suggests that the latent states identified by the HMM are economically meaningful. In other words, the estimated regimes are not arbitrary statistical clusters, but correspond to recognizable differences in market risk conditions.

The bottom panel reports the log-likelihood of the rolling HMM estimation. Although the log-likelihood varies over time, the model remains capable of adapting to changing market structures through regular re-estimation. This rolling framework is especially appropriate for financial data, where relationships among volatility, credit conditions, and market stress are not stable over very long horizons.

Overall, the figure suggests that the rolling HMM provides a useful latent-state representation of market conditions. The identified regimes align well with major historical stress episodes and generate interpretable probability signals that can be incorporated into the downstream crash prediction model as regime-aware explanatory features.

Feature_Pattern	Description
*_logret	Daily log return: ln(P_t / P_{t-1})
*_diff	Daily difference: P_t - P_{t-1}, used for rates and VIX
*_range	Daily range: log(High / Low), measures intraday volatility
RATE_SLOPE_10Y_3M	Interest rate slope: 10Y yield - 3M yield, yield curve steepness
SPX_VOL_20D	S&P 500 20-day rolling volatility (annualized)
SPX_MAXDD_20D	S&P 500 20-day rolling maximum drawdown
CREDIT_FACTOR	Credit spread factor: r_HYG - r_IEF, high yield vs treasury return
*_5d	5-day rolling mean (min_periods=4)
*_10d	10-day rolling mean (min_periods=8)
*_q252	252-day rolling 90th percentile (min_periods=202)
SPREAD_VIX_HIGH	spread_gate × vix_gate, where gates map values to 0~1 based on 252d rolling 80th-95th percentile
SPX_VIX_SPIKE	(-SPX_ret_1) × max(dVIX_1, 0), captures SPX loss × VIX spike interaction
DXY_SPREAD_INTERACT	DXY_ret_20 × spread_z, where spread_z = (spread - rolling_mean) / rolling_std (252d)
VIX_VOL_RATIO	VIX Close / SPX_VOL_20D, ratio of implied volatility to realized volatility
SPX_VOL_STRESS	SPX_VOL_20D_zscore_252 × P_Stress, volatility z-score interaction with stress probability
HYG_RANGE_STRESS	HYG_range_5d_zscore_252 × P_Stress, HYG range z-score interaction with stress probability
NASDAQ_RANGE_STRESS	NASDAQ100_range_zscore_252 × P_Stress, NASDAQ range z-score interaction with stress probability
*_RR_60d	Recurrence Rate from 60-day window recurrence plot, ε = 0.5 × std
*_DET_60d	Determinism from 60-day window recurrence plot, proportion of recurrence points in diagonal lines

Table 2 Feature Definition

Category	Feature_Count
Credit Spread	1
Determinism (60d)	4
Difference	9
Interaction (DXY×Spread)	1
Interaction (SPX×VIX)	1
Interaction (Spread×VIX)	1
Interaction (Z-Score×P_Stress)	2
Interest Rate Slope	4
Log Return	48
Max Drawdown	4
Range (log H/L)	60
Recurrence Rate (60d)	4
Rolling 10D Mean	2
Rolling 252D Q90	2
Rolling 5D Mean	4
Vol Ratio (Implied/Realized)	1
Volatility	5
total	**153**

Table 3 Feature Number

Dimensionality reduction and feature selection

After the raw feature set is generated, the project performs an additional feature selection step. Importantly, all filtering and ranking are conducted on the training period only, using fixed split boundaries, in order to avoid leakage from validation and test sets.

First, a correlation analysis is performed on numeric training features, and highly correlated feature pairs with $|\ corr\ | > 0.8$ are identified. For each of such pair, the feature with the shorter effective history is removed, while the longer series is retained. This helps reduce redundancy without sacrificing data coverage.

Second, the remaining features are evaluated one by one using ROC-AUC and PR-AUC against the crash labels. Because crash prediction is an imbalanced classification task, PR-AUC is used as the main ranking criterion. For each feature, the code also evaluates the sign-reversed version and keeps whichever direction yields better discrimination. This allows the screening step to focus on predictive strength rather than being constrained by a prespecified sign assumption.

Metric	Value
Total Features (Original)	156
Features After Removing High Correlation	74
Features Removed	82
Highly Correlated Pairs (|corr|>0.8)	284
Training Samples	9959
Training Date Range Start	1980-01-02
Training Date Range End	2019-06-28
SP500 - Best PR-AUC Feature	CREDIT_FACTOR_RR_60d
SP500 - Best PR-AUC Score	0.257
SP500 - Best ROC-AUC Feature	IEF_range_5d
SP500 - Best ROC-AUC Score	0.7471

Table 4 Summary of Feature Selection

7. Data Mining Models / Methods

Feature Groups for Model Comparison

To evaluate the predictive contribution of different feature sets, four feature groups are constructed:

Group 1 – Selected Financial Indicators

A curated (manually selected) set of 14 economically motivated features including: Credit market indicators, Volatility signals, Cross-market interactions, Regime information. This group represents the core feature set designed for crash prediction.

Group 2 – Regime Features Only

This group includes only regime-related variables: P_{Calm}, $P_{Transitional}$, P_{Stress}, $\Delta P_{Stress}^{(5)}$. The purpose of this group is to evaluate whether market regime information alone can provide predictive power.

Group 3 – Financial Indicators without Regime Signals

This group is identical to Group 1 but excludes regime-related features. Comparing Group 1 and Group 3 allows us to measure the incremental predictive value of regime detection.

Group 4 – Top 20 Data-driven Features

The fourth group consists of the top 20 features ranked by feature importance, selected from the prior step in feature selection. Feature-specific preprocessing rules are automatically applied to this group based on the dynamic scaling rules.

Group	Feature_Index	Feature	Group	Feature_Index	Feature
Group1_14Features	1	CREDIT_FACTOR_RR_60d	Group2_RegimeProb	1	P_Calm
Group1_14Features	2	SPX_VOL_STRESS	Group2_RegimeProb	2	P_Transitional
Group1_14Features	3	EUROSTOXX50_range_z252	Group2_RegimeProb	3	P_Stress
Group1_14Features	4	DXY_SPREAD_INTERACT	Group2_RegimeProb	4	Delta_P_Stress_5d_z252
Group1_14Features	5	COMMODITY_range_z252			
Group1_14Features	6	VIX_RR_60d	Group3_NoRegime	1	CREDIT_FACTOR_RR_60d
Group1_14Features	7	HYG_range_z252	Group3_NoRegime	2	EUROSTOXX50_range_z252
Group1_14Features	8	HYG_RANGE_STRESS	Group3_NoRegime	3	DXY_SPREAD_INTERACT
Group1_14Features	9	HYG_logret	Group3_NoRegime	4	COMMODITY_range_z252
Group1_14Features	10	SP500_range_z252	Group3_NoRegime	5	VIX_RR_60d
Group1_14Features	11	VIX_VOL_RATIO	Group3_NoRegime	6	HYG_range_z252
Group1_14Features	12	P_Stress	Group3_NoRegime	7	HYG_logret
Group1_14Features	13	IEF_range_5d_z252	Group3_NoRegime	8	SP500_range_z252
Group1_14Features	14	NASDAQ_RANGE_STRESS	Group3_NoRegime	9	VIX_VOL_RATIO
			Group3_NoRegime	10	IEF_range_5d_z252

Table 5 Feature Selection for Group 1-Group 3

Feature	ROC_AUC	PR_AUC	Direction	Valid_Samples	Positive_Rate	Rank_PR_AUC	Rank_ROC_AUC
CREDIT_FACTOR_RR_60d	0.658712435	0.257031375	Positive	3029	0.040277319	1	12
HYG_range	0.709235527	0.15912367	Positive	3077	0.039649009	2	6
IEF_range_5d	0.747102399	0.139273312	Positive	4256	0.033599624	3	1
HYG_logret	0.531670033	0.110717617	Positive	3076	0.039661899	4	44
EUROSTOXX50_range	0.663050276	0.099987042	Positive	3084	0.039559014	5	10
IEF_range	0.710664336	0.094286605	Positive	4259	0.033575957	6	5
HYG_logret_10d	0.591173575	0.091704398	Negative	3069	0.039752362	7	24
VIX_RR_60d	0.580149377	0.083777732	Negative	7383	0.031017202	8	29
HYG_logret_5d	0.530267595	0.082192626	Negative	3073	0.039700618	9	46
COMMODITY_range	0.600589298	0.081616828	Positive	8947	0.02906002	10	21
DXY_SPREAD_INTERACT	0.502346052	0.081240853	Positive	2875	0.038956522	11	72
SP500_range	0.720262114	0.081188214	Positive	9959	0.029521036	12	4
SPX_MAXDD_20D_q252	0.734771825	0.07657741	Negative	9728	0.028474507	13	2
USDJPY_range_5d	0.63080775	0.07314696	Positive	5700	0.038070175	14	15
NASDAQ100_range	0.706798168	0.070368866	Positive	8506	0.030566659	15	7
VIX_VOL_RATIO	0.643391053	0.068871202	Negative	7431	0.031085991	16	13
RATE_10Y_diff_10d	0.51971095	0.068706961	Positive	9951	0.029544769	17	53
EUROSTOXX50_logret_5d	0.578966182	0.064923393	Negative	3080	0.03961039	18	30
VIX_diff	0.5189218	0.063512488	Negative	7430	0.030955585	19	55
SPREAD_VIX_HIGH	0.537696797	0.061772063	Positive	2875	0.038956522	20	42

Table 6 Top 20 Features for SP500

Dataset Split

To ensure realistic out-of-sample evaluation, the dataset is split chronologically. The test set remains fixed throughout the experiments. The remaining data is divided into training and validation sets using a 2:1 chronological split. This time-based split avoids information leakage that could occur with random cross-validation in financial time series.

Models Evaluated

Three classification algorithms are compared for every group of features:

Logistic Regression with L1 regularization. The L1 penalty encourages feature sparsity and improves interpretability.

Random Forest is a tree-based ensemble method that captures nonlinear relationships.**,** This model is robust to noisy financial features and complex interactions. Key parameters tuned using grid search: Number of trees, Maximum tree depth, Minimum sample splits, and Class weighting

LightGBM is a gradient boosting decision tree model optimized for efficiency. Gradient boosting often performs well on structured tabular data with nonlinear dependencies. Hyperparameters tuned include: Number of trees, Learning rate, Tree depth, Number of leaves and Class imbalance weighting.

Model Selection Procedure

The model training workflow follows a three-stage evaluation process:

1. Hyperparameters are tuned using the training set.
2. The best-performing model for each feature group is then retrained using the combined training and validation data before being evaluated on the test set.

3. Final best-performing model across all four groups evaluated on test set is compared with baseline model (a random ranking method).

This procedure ensures that the test set remains completely unseen during model selection.

Evaluation Metrics

Because crash events are rare, evaluation focuses on metrics suitable for imbalanced classification problems. Primary metrics include: ROC-AUC, which measures the model's ability to distinguish crash and non-crash periods; PR-AUC, Precision-Recall AUC is particularly informative when positive events are rare; Recall@10%, which measures the proportion of crashes captured within the top 10% highest predicted risk days.: Precision@10%, which measures the accuracy of predictions within the highest risk segment.

Baseline performance provides a reference point to evaluate the predictive value of the machine learning models. Expected baseline performance:

$ROC\text{-}AUC \approx 0.5$, $PR\text{-}AUC \approx$ crash frequency.

8. Performance Evaluation

Group	Num_Features	Train_Samples	Val_Samples	Test_Samples	Top10%_Count	Test_Positive_Rate	Test_Total_Crashes
Group1_14Features	14	2426	1213	892	89	0.048206278	43
Group2_RegimeProb	4	2292	1146	892	89	0.048206278	43
Group3_NoRegime	10	2427	1214	892	89	0.048206278	43
Group4_Top20	20	2427	1214	892	89	0.048206278	43

Group	LR_Test_PR_AUC	RF_Test_PR_AUC	LGB_Test_PR_AUC	LR_Val_PR_AUC	RF_Val_PR_AUC	LGB_Val_PR_AUC	LR_Test_ROC_AUC
Group1_14Features	0.041399157	0.045844266	0.050175113	0.116241852	0.128634617	0.107279908	0.365080669
Group2_RegimeProb	0.049300186	0.04947737	0.049368779	0.166617715	0.079011323	0.100685703	0.514284932
Group3_NoRegime	0.043553962	0.050887074	0.052457582	0.089465815	0.104706236	0.097276182	0.426247021
Group4_Top20	0.057494594	**0.092291805**	0.084540018	0.115350101	0.168234049	0.176359449	0.52258471

Group	RF_Test_ROC_AUC	LGB_Test_ROC_AUC
Group1_14Features	0.438299504	0.487385981
Group2_RegimeProb	0.478250746	0.477716602
Group3_NoRegime	0.53937601	0.52189991
Group4_Top20	0.586709398	0.562138768

Group	LR_TP@10%	LR_Recall@10%	LR_Precision@10%	RF_TP@10%	RF_Recall@10%	RF_Precision@10%	LGB_TP@10%
Group1_14Features	5	0.11627907	0.056179775	3	0.069767442	0.033707865	3
Group2_RegimeProb	5	0.11627907	0.056179775	5	0.11627907	0.056179775	1
Group3_NoRegime	3	0.069767442	0.033707865	2	0.046511628	0.02247191	3
Group4_Top20	5	0.11627907	0.056179775	14	0.325581395	0.157303371	10

Group	LGB_Recall@10%	LGB_Precision@10%
Group1_14Features	0.069767442	0.033707865
Group2_RegimeProb	0.023255814	0.011235955
Group3_NoRegime	0.069767442	0.033707865
Group4_Top20	0.23255814	0.112359551

Group	Best Model	Test_ROC_AUC	Test_PR_AUC	Top10%_Count	TP@10%	Recall@10%	Precision@10%
Baseline	RandomRank	0.5	0.048206278	89	4.3	0.1	0.048206278
Group1_14Features	Random Forest	0.438299504	0.045844266	89	3	0.069767442	0.033707865
Group2_RegimeProb	L1 Logistic	0.514284932	0.049300186	89	5	0.11627907	0.056179775
Group3_NoRegime	Random Forest	0.53937601	0.050887074	89	2	0.046511628	0.02247191
Group4_Top20	LightGBM	0.562138768	0.084540018	89	10	0.23255814	0.112359551

Table 7 Report on All Model Performance

From table 7, it shows that Group 4 (Top 20 features) achieved the strongest predictive performance among all feature sets. Although LightGBM achieved the highest validation

PR-AUC within Group 4 (0.1764) and was therefore selected as the official best model under the validation-based selection rule, the Random Forest model actually delivered the strongest out-of-sample test performance, with ROC-AUC = 0.5867, PR-AUC = 0.0923, Recall@10% = 0.3256, and Precision@10% = 0.1573. This means that among the top 10% highest-risk predicted days, the Random Forest model correctly identified 14 out of 43 crash days, substantially outperforming the random baseline.

By contrast, the selected LightGBM model in Group 4 still showed strong test performance, with ROC-AUC = 0.5621 and PR-AUC = 0.0845, clearly above the random-ranking baseline (PR-AUC = 0.0482). However, its performance was slightly weaker than Random Forest on the test set, suggesting that the validation winner did not fully generalize as the best out-of-sample model.

Group	Model	Test_Samples	Test_Positive_Rate	Test_ROC_AUC	Test_PR_AUC	Top10%_Count	TP@10%	Recall@10%	Precision@10%
Group4_Top20	Random Forest	892	0.048206278	0.586709398	0.092291805	89	14	0.325581395	0.157303371
Group4_Top20	Baseline_RandomRank	892	0.048206278	0.5	0.048206278	89	4.3	0.1	0.048206278
Group4_Top20	Ratio (Model/Baseline)	0	0	1.173418796	1.914518365	0	3.25581	3.255813953	3.263130389

Table 8 Performance Comparison of Best-Performing Model (Random Forest) and Baseline

Group 2, which used only regime probability features, produced relatively stable but modest predictive performance. Its selected L1 Logistic Regression model achieved ROC-AUC = 0.5143 and PR-AUC = 0.0493, only slightly above baseline. This suggests that regime information alone contains useful directional information, but its standalone predictive power is limited.

Group 1 and Group 3 showed weaker performance overall. In particular, Group 1 underperformed the baseline on several test metrics, indicating that the manually selected 14-feature set may not generalize well to the test period. Group 3, which removed regime-related features from Group 1, achieved slightly better ROC-AUC than Group 1 but still weak PR-AUC, suggesting that the manually curated feature subset was not sufficient for strong crash prediction.

These results imply that feature richness and data-driven feature ranking are more important than relying only on a small hand-picked set of indicators. At the same time, regime features appear to add useful context, but their value is most evident when combined with other market indicators rather than used alone.

9. Project Results

This section provides a further evaluation of the Random Forest model with Group 4 features, focusing on both predictive performance and economic interpretation.

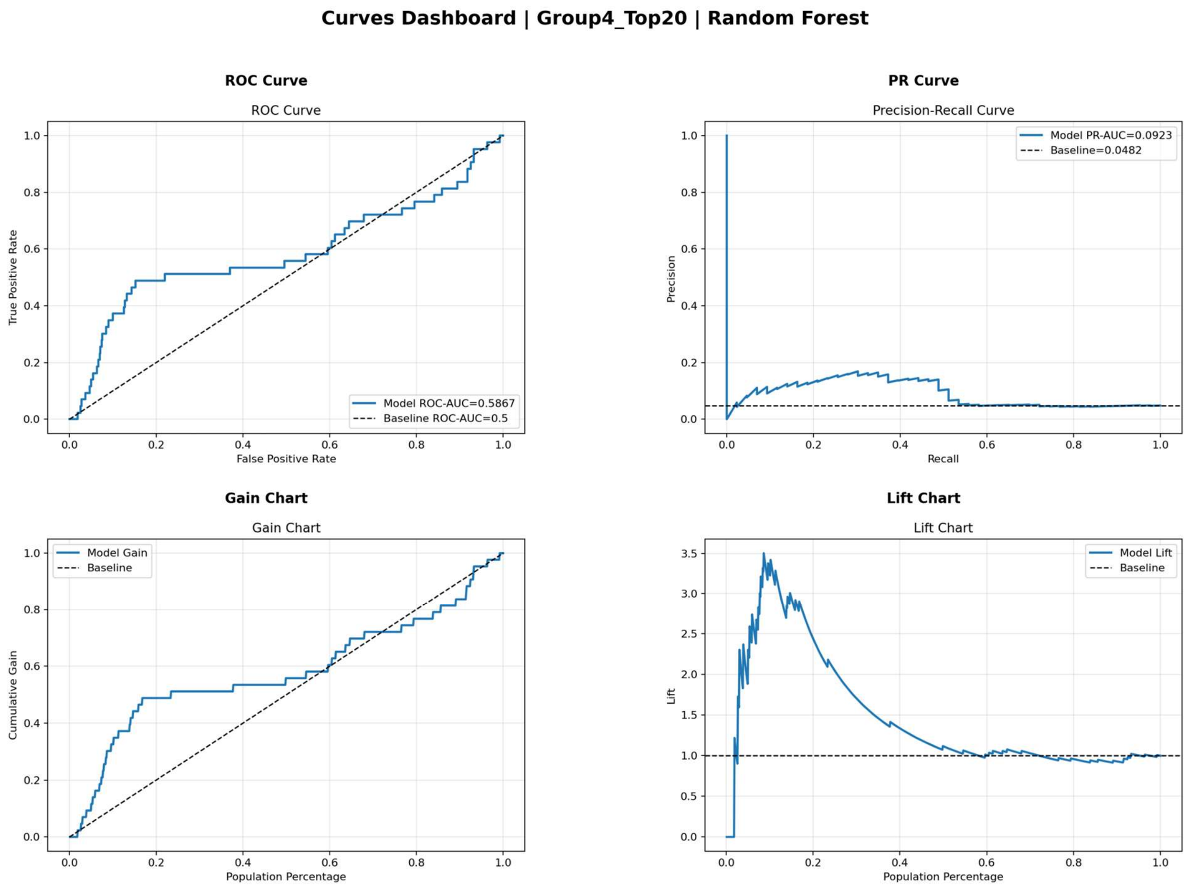

Figure 10 AUROC, AUPRC, Gain Chart and Lift Chart of Performance

ROC, PR curve, gain chart and lift chart are shown in figure 10. These plots show that the selected model performs better than a random baseline in identifying future crash events. In particular, the gain and lift curves indicate that the model is able to concentrate a larger proportion of true crash days within the highest-risk predictions, which is important because crash events are rare and standard accuracy alone is not informative in this setting. This result suggests that the model has practical early-warning ability rather than merely fitting noise.

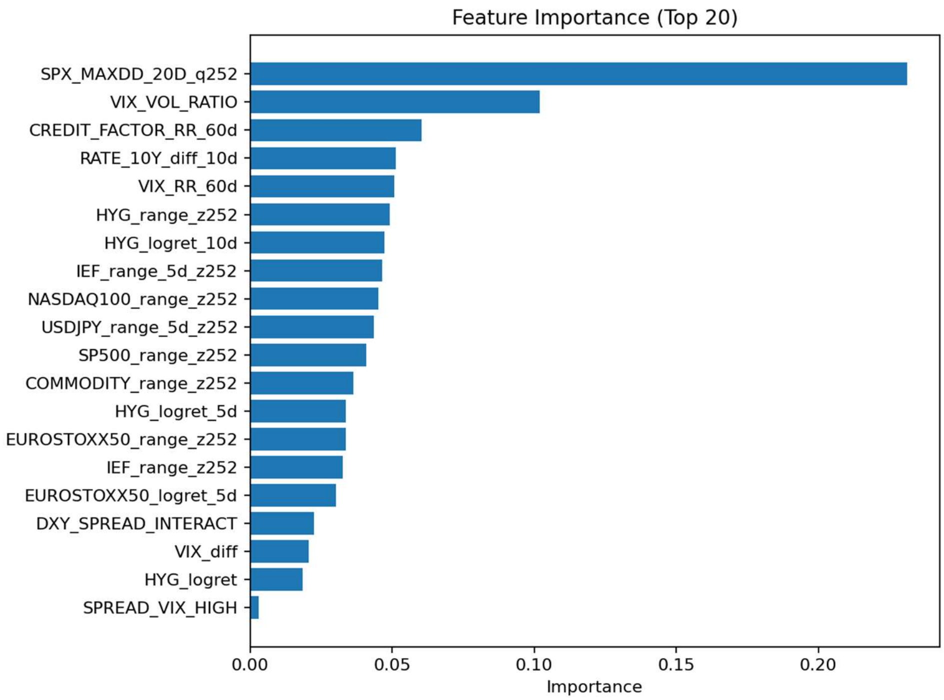

Figure 11 Feature Importance

Figure 11 shows the conventional feature importance ranking from the tree-based model. The most important variables are related to recent equity drawdowns, volatility conditions, credit market behavior, and interest-rate changes. This indicates that market crash risk is not driven by a single signal, but instead emerges from the joint interaction of several macro-financial channels. In other words, the model captures a multi-factor risk structure involving equity stress, volatility regime shifts, credit conditions, and macro rate dynamics.

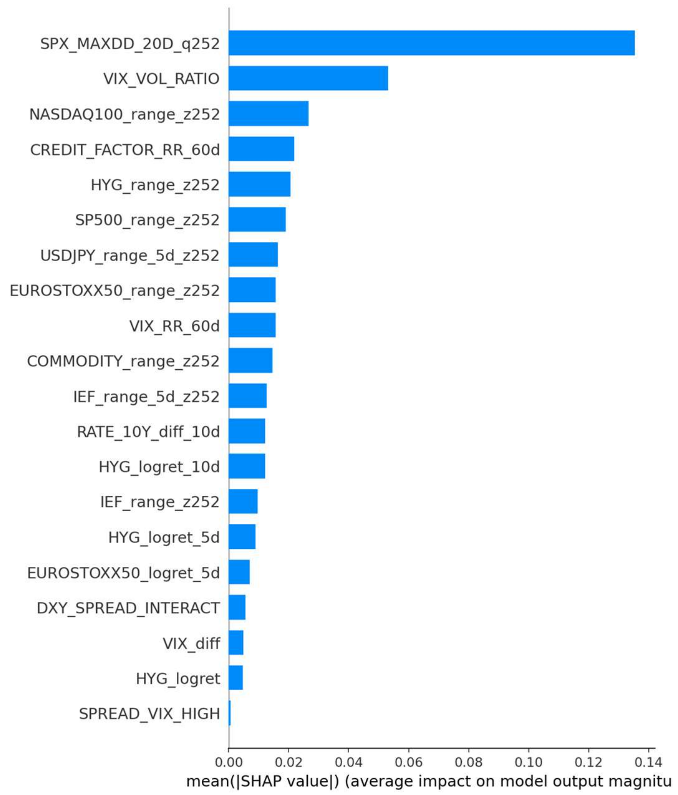

Figure 12 Mean SHAP Value of Features

The SHAP bar plot presents the global importance ranking of the top features based on their mean absolute SHAP values. Unlike conventional feature importance measures derived from tree splits, this figure reflects the average magnitude of each feature's contribution to the model output across all observations, regardless of direction. The plot shows that SPX_MAXDD_20D_q252 is by far the most influential feature in the Random Forest model, followed by VIX_VOL_RATIO, NASDAQ100_range_z252, and CREDIT_FACTOR_RR_60d. This ranking suggests that the model relies primarily on recent market drawdown conditions, volatility regime information, technology-sector range expansion, and credit-market structural signals when predicting future crash risk. The large gap between the first feature and the others also indicates that recent S&P 500 drawdown carries especially strong explanatory power for identifying high-risk market states. Overall, the figure confirms that the model's predictions are driven by a combination of equity stress, volatility conditions, cross-asset dispersion, and credit-market dynamics.

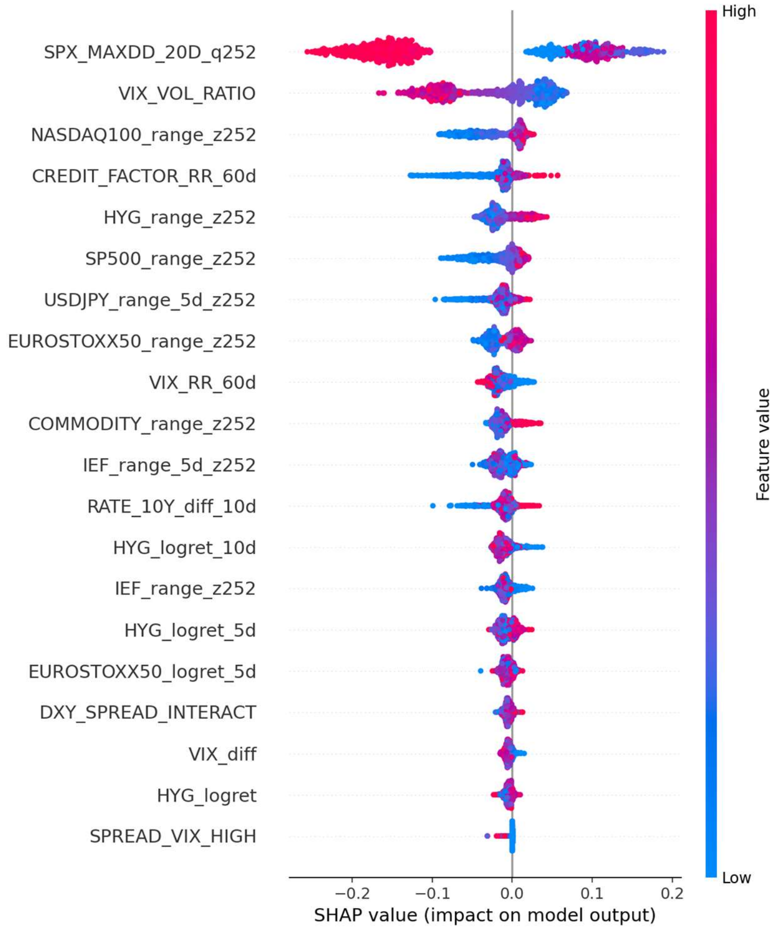

Figure 13 SHAP values of Features

The SHAP summary plot provides a global view of how the most important features influence the Random Forest model's crash predictions across all test observations. The horizontal spread of each feature reflects the magnitude of its contribution to the model output, while the color scale shows whether the feature value is relatively high or low. The plot indicates that SPX_MAXDD_20D_q252 is the most influential feature, followed by VIX_VOL_RATIO, NASDAQ100_range_z252, and CREDIT_FACTOR_RR_60d. The color distribution suggests that deeper recent drawdowns in the S&P 500 tend to increase predicted crash risk, while larger values of VIX_VOL_RATIO tend to reduce it. In addition, higher values of NASDAQ100_range_z252 and CREDIT_FACTOR_RR_60d are generally associated with positive SHAP contributions. Overall, the figure suggests that the model captures crash risk through a combination of recent equity stress, volatility regime conditions, technology-sector instability, and credit-market structure.

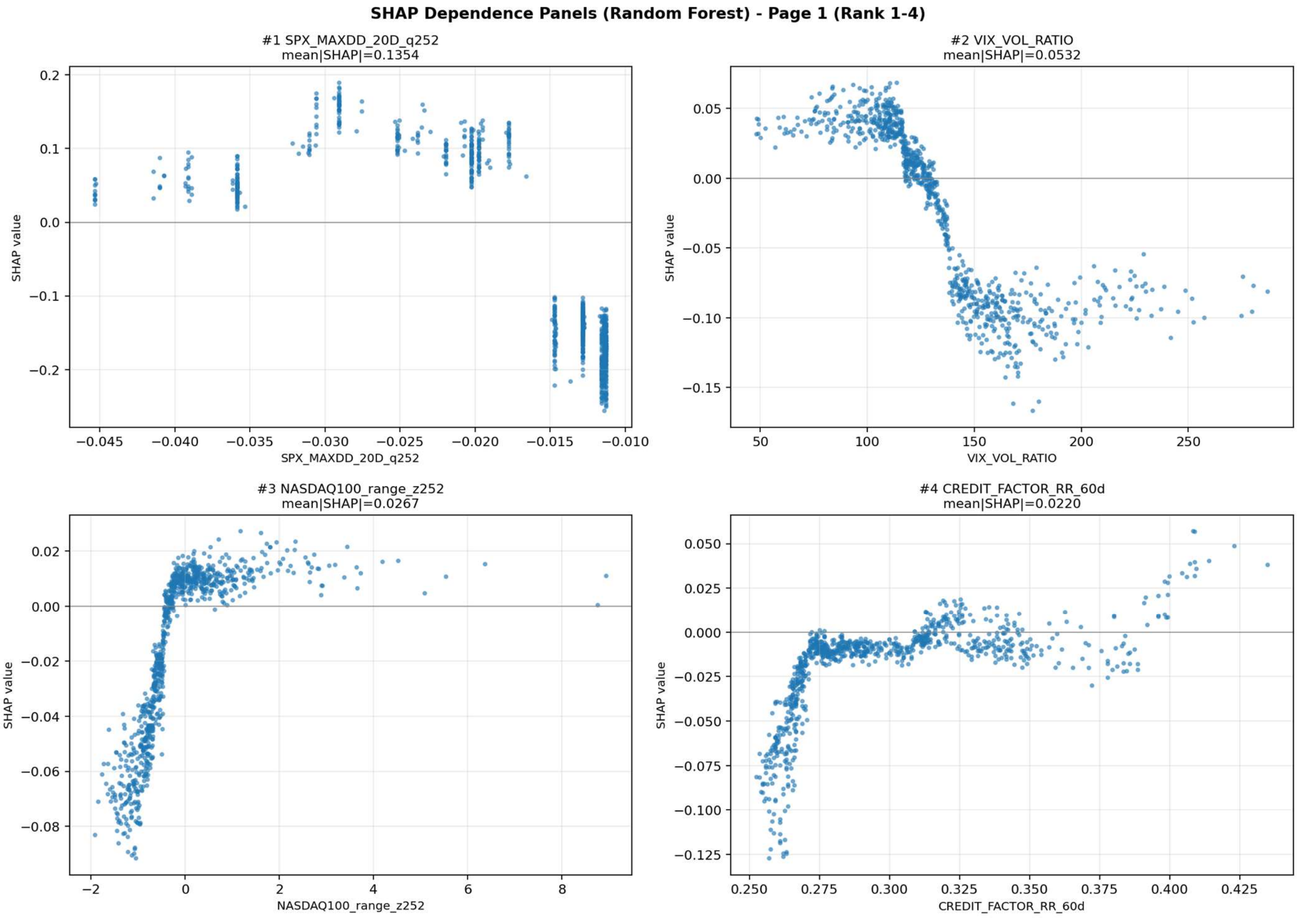

Figure 14 SHAP Dependence Analysis

The SHAP dependence panels provide a more detailed interpretation of the top-ranked features by showing how each feature value is related to its SHAP contribution. Unlike the summary plot, which emphasizes global importance, these panels reveal the functional shape of each relationship, including nonlinearity and threshold effects. The panel for SPX_MAXDD_20D_q252 shows that deeper recent drawdowns strongly increase predicted crash risk, while milder drawdowns reduce it. VIX_VOL_RATIO exhibits a clear negative relationship with SHAP values, indicating that lower volatility-ratio regimes are associated with higher future crash risk, whereas higher values tend to lower the prediction. NASDAQ100_range_z252 shows an upward relationship, suggesting that a wider Nasdaq trading range is linked to rising market fragility. Finally, CREDIT_FACTOR_RR_60d displays an overall positive association with SHAP values, implying that stronger recurrence structure in credit-market dynamics may signal more persistent systemic stress. Together, these dependence panels show that the model learns economically meaningful and nonlinear relationships rather than relying on simple linear effects.

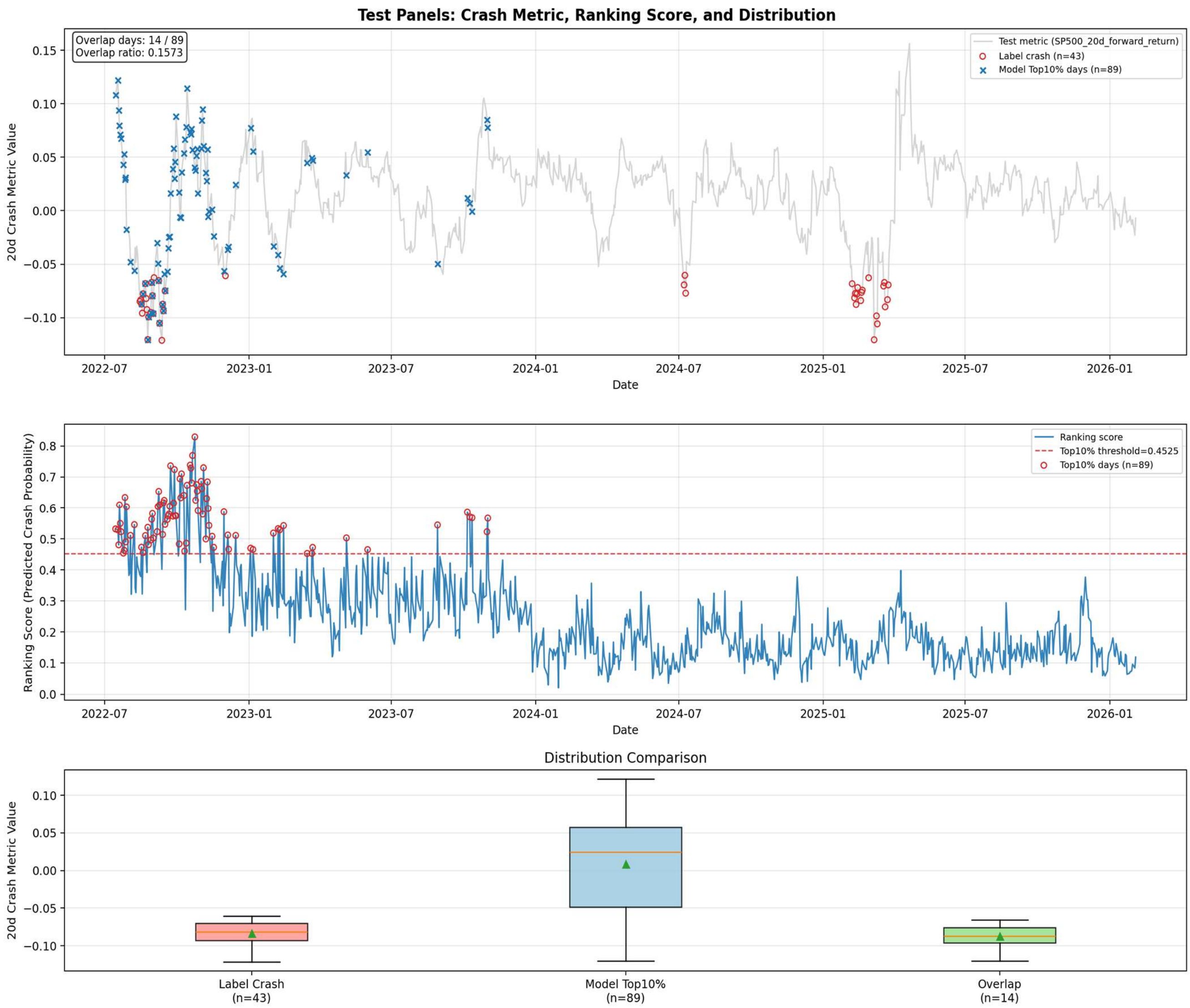

Figure 15 Crash Days Timeline and Distribution (Forecast V.S. Reality)

Figure 15 provides an integrated evaluation of the model's behavior on the test set by combining three complementary views: the realized crash metric over time, the model's ranking score timeline, and the distributional comparison between labeled crash days and model-selected high-risk days.

The top panel compares the realized 20-day forward return metric with two event sets: the true labeled crash days and the model-selected top 10% highest-risk days. The figure shows that the model identifies some severe crash episodes, particularly in the earlier part of the test period, but the overlap is limited. Specifically, 14 out of 89 model-selected top-risk days overlap with the 43 labeled crash days, corresponding to an overlap ratio of 15.73%. This indicates that the model captures part of the true crash structure, but also selects many days that are not ultimately labeled as crashes.

The middle panel shows the time series of the model's ranking scores together with the global top-10% threshold. A clear temporal concentration is visible: most of the highest model scores occur in the early portion of the test set, especially from mid-2022 to late 2023. After that, the predicted scores decline substantially and remain mostly below the global top-10% cutoff. This suggests that the model continues to assign nonzero risk throughout the test period, but its strongest signals remain concentrated in earlier regimes. Consequently, later labeled crash events are not included in the global top-risk set because their scores are relatively low compared with those assigned in the earlier period.

The bottom panel compares the distributions of the crash metric for three groups: all labeled crash days, the model-selected top-10% days, and the overlap set. The labeled crash days have strongly negative metric values, as expected from the crash definition. In contrast, the model top-10% group exhibits a much wider distribution and includes many observations with weakly negative or even positive values, showing that not all highly ranked model signals correspond to realized crashes. However, the overlap group has a distribution close to that of the labeled crash days, suggesting that when the model does correctly identify crash periods, it captures economically meaningful extreme downside events.

Overall, Figure 15 highlights both the strengths and the limitations of the model. On the positive side, the model is able to rank a subset of severe crash periods highly, and the overlap set aligns well with the true crash distribution. On the negative side, its high-risk predictions are heavily concentrated in the early part of the test sample. This pattern suggests a form of regime change or concept drift: in the earlier period, crash risk was more strongly associated with the feature patterns learned from the training data, such as recent equity drawdowns, volatility-regime signals, and credit-related stress structure. In the later period, however, realized crash events appear to have emerged through different market mechanisms, so that the same feature combinations no longer mapped to the same future downside outcomes. As a result, the model continued to detect states resembling the older stress regime, but those states were no longer the ones most closely associated with subsequent crashes. Therefore, the figure suggests that the model has useful early-warning value, but its ranking mechanism is temporally unstable out of sample and sensitive to changes in market structure over time.

10. Impact of the Project Outcomes

This project demonstrates that a regime-aware machine learning framework can provide meaningful value for market risk monitoring and early warning. By combining rolling Hidden Markov Model (HMM) regime identification with supervised crash prediction, the study shows that multi-asset market information contains useful signals for identifying periods of elevated downside risk. The results suggest that recent equity drawdowns, volatility regime indicators, and credit-related features are all economically meaningful contributors to crash prediction, which supports the idea that market stress is driven by the interaction of multiple macro-financial channels rather than by any single variable.

From a practical perspective, the project shows that the model is able to identify a subset of severe crash periods and assign them relatively high-risk scores. This indicates that the framework has potential value as an early-warning tool for investors, portfolio managers, and risk-monitoring systems. Even though the model does not perfectly capture all crash events, it provides a structured way to rank periods by estimated risk and to focus attention on the most vulnerable market states. In this sense, the project contributes not only a predictive model, but also an interpretable monitoring framework that links statistical learning results to economically meaningful market conditions.

At the same time, the study also reveals important limitations. The test-period analysis shows that the model's strongest predictions are concentrated in earlier parts of the sample, while later crash events are not captured as effectively. This suggests that the relationship between predictors and future crash outcomes is not stable over time. In other words, market structure may evolve, and crash mechanisms in later periods may differ from those represented in the training data. This finding is important in itself, because it shows that building an effective financial early-warning system is not only a modeling problem, but also a regime-adaptation problem. A model that performs well in one market environment may lose effectiveness when the underlying structure of risk changes.

Therefore, one of the main impacts of this project is that it highlights both the usefulness and the limits of static machine learning models in financial prediction. The results support the use of machine learning for market risk assessment, but they also emphasize the need for adaptive modeling strategies. Future work could improve the framework by introducing rolling retraining, dynamic thresholds, time-varying calibration, and more explicit treatment of concept drift. It would also be valuable to expand the feature space with additional macroeconomic, liquidity, and cross-sectional market indicators, or to explore sequence-based models that may better capture evolving market dynamics.

Overall, the outcomes of this project are valuable in three ways. First, they confirm that multi-asset regime-aware features can improve the understanding and prediction of crash risk. Second, they provide interpretable evidence that connects model outputs to financially meaningful variables such as drawdowns, volatility regimes, and credit stress. Third, they reveal that temporal instability and regime change are central challenges for out-of-sample crash prediction. For these reasons, the project not only produces a useful predictive framework, but also offers important insight into how future financial risk models should be designed, tested, and updated in changing market environments.